Supporting information

Rescue of the Orphan Enzyme Isoguanine Deaminase

Daniel S. Hitchcock, Alexander A. Fedorov, Elena V. Fedorov, Lawrence J. Dangott,

Steven C. Almo, and Frank M. Raushel

Crystallization and Data Collection.

The crystals of cytosine deaminase from *Escherichia coli* complexed with Zn^{2+} were grown by hanging drop vapor diffusion at room temperature. The protein solution contained wild-type CDA (18 mg/mL) in 50 mM Tris (pH 7.5), and 1 mM $ZnCl_2$; the precipitant contained 35% pentaerythritol propoxylate, 0.05 M Hepes (pH 7.5), 0.2 M potassium chloride, and 1 mM $ZnCl_2$. The crystals of the enzyme appeared in 3-4 days and exhibited diffraction consistent with the space group R32, with one molecule of the protein per asymmetric unit (**Table S1**).

Isoguanine was insoluble in the number of different solvents; therefore, the powder of the isoguanine was used directly for soaking experiments. Crystals containing CDA.Zn^{2+} were soaked for 6 days in mother liquor and excess of the isoguanine powder. Prior to data collection, crystals were transferred to cryoprotectant solutions composed of their mother liquids supplemented with 20% glycerol. After incubation for ~10 s, the crystals were flash-cooled in a nitrogen stream. Diffraction data were collected at the NSLS X4A beamline (Brookhaven National Laboratory) on an ADSC CCD detector. Diffraction intensities were integrated and scaled with programs DENZO and SCALEPACK *(1)*. The data collection statistics are given in **Table S1**.

Structure Determination and Model Refinement.

The structure of CDA crystalline complex was determined by molecular replacement with the fully automated molecular replacement pipeline BALBES *(2)*, using only input amplitudes, sequence data and the *E. coli* CDA (PDB code: 1K6W) as the search model. A partially refined structure of CDA complex was the output from BALBES and several iterative cycles of refinement were performed for the CDA ternary complex, including model rebuilding with COOT *(3)*, refinement with PHENIX *(4)*, and automatic model rebuilding with ARP *(5)*. Final crystallographic refinement statistics for the CDA·Zn^{2+}·Isoguanine complex are provided in the **Table S1**.

The model of the CDA·Zn^{2+}·Isoguanine complex was refined at 2.25 Å with an R_{cryst} of 0.207 and an R_{free} of 0.243. The final structure contains protein residues 4-425, two well-defined Zn^{2+} ions, and one isoguanine molecule bound in the CDA active site. The first Zn^{2+} ion is bound in the active site and the second Zn^{2+} is bound between two symmetry related molecules.

References

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Supplemental Table S1. Purification Scheme for Isoguanine Deaminase[a]

Step	volume (mL)	units (U)	protein (mg)	U/mg
lysate	75	8.6	490	0.018
protamine sulfate	147	6.2	130	0.048
$(NH_4)_2SO_4$	5	5.4	10	0.53
gel filtration	17	2.2	0.63	3.4
anion exchange	4.5	0.97	0.11	14

[a]A Unit (U) is defined as 1.0 μmol/min.

Supplemental Table S2. Fragments from nanoLC Electrospray MSMS for Cytosine Deaminase[a]

Position	Observed mass	Actual Mass
3-14	1317.26 (+2)	1315.25
15-30	1822.05 (+3)	1819.02
31-46	1647.20 (+2)	1645.18
95-103	1012.60 (+2)	1010.59
112-122	1323.68 (+2)	1321.67
193-201	1062.06 (+2)	1060.04
227-236	1059.00 (+2)	1056.96
244-262	2058.06 (+3)	2055.04
375-391	1817.46 (+3)	1814.43
405-427	2570.04 (+3)	2567.02

[a]The position indicates the original location of the indicated peptide in the intact protein after cleavage by trypsin. The charge of the fragment is shown next to the observed mass.

Supplemental Table S3. Data collection and refinement statistics for crystals of the CDA·Zn^{2+} · Isoguanine complex.

	CDA·Zn^{2+}·Isoguanine
Data collection	
Space group	R32
No. of mol. in asym. unit	1
Cell dimensions	
a, *c* (Å)	147.187, 199.747
Resolution (Å)	2.25
No. of unique reflections	36490
R_{merge}	0.087
$I / \sigma I$	7.9
Completeness (%)	92.8
Refinement	
Resolution (Å)	39.0-2.25
R_{cryst}	0.207
R_{free}	0.243
R.m.s. deviations	
Bond lengths (Å)	0.008
Bond angles (º)	1.09
No. atoms	
Protein	3336
Waters	80
Inhibitor	11
Zn^{2+} ions	2
PDB entry	3RN6